5.3 Properties of Logarithms

Name _____ Period ____ Date _____

Pre-Calculus 5.3 (30-minute plan) · Student sheet

I can…

- use the change-of-base formula.
- expand a logarithmic expression.
- condense a logarithmic expression.

Vocabulary · fill in the missing word

Change-of-base formula — $\log_a x =$ _____ $= \frac{\ln x}{\ln a}$

Expand — rewrite a logarithm of a product, quotient, or power as a sum, _____, or multiple of logarithms

Condense — rewrite a sum or _____ of logarithms as a single logarithm

Part 1 · Notes and examples (fill in each note, then work the example)

Change of base: $\log_a x =$ _____ (or log x / log a)

Example 1

Use a calculator. Round to four decimal places.

a. $\log_4 25$

b. $\log_2 12$

Product and quotient: $\log_a(uv) =$ _____; $\log_a(u/v) = \log_a u - \log_a v$

Power: $\log_a u^n =$ _____

Example 2

Expand.

a. $\log_4 5x^3y$

b. $\ln\left(\frac{\sqrt{3x-5}}{7}\right)$

Example 3

Condense.

a. $\frac{1}{2}\log x + 3\log(x + 1)$

b. $2\ln(x + 2) - \ln x$

Part 2 · Partner practice

Partner practice

Condense $\log 4 + 2\log x - \log 3$.